CP020003 — Artificial Intelligence 2026

Khon Kaen University

Week 4 — Homework Assignment (DNA)

This week we move from the synthetic **DNA-classification dataset** used in class into a real epigenetics problem: DNA methylation. Methylation is a chemical mark — a methyl group attached to a cytosine, almost always right before a guanine, forming a "CpG site" — that turns genes up or down without changing the underlying DNA sequence. Which CpG sites are methylated, and how strongly, is itself something that can be predicted from local sequence and genomic context, and that is exactly the modeling task in front of you this week.

Your job is to take the same supervised-learning workflow we practiced together in Week4— EDA, feature engineering (GC content, k-mer frequencies), classical ML (Decision Tree, Random Forest, XGBoost), and PyTorch deep learning directly on raw sequence (CNN, LSTM, GRU) — and apply it to this new, real dataset.

1. Dataset

Download and unzip the dataset from the course GitHub repository:

```
https://github.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1/
blob/main/dataset/dna-methylation.zip
```

After unzipping, you will see the following structure:

```
dna-methylation/
    train.csv
    test.csv
```

train.csv contains the Beta target column described below; test.csv contains the same feature columns but withholds Beta — that is what your trained model will predict.

Data Dictionary

Column	Description
`Id`	Unique CpG probe identifier (e.g. cg00007981) — the Illumina 450K methylation array probe name.
`CHR`	Chromosome number where the CpG site is located (1–22, X, Y).
`MAPINFO`	Genomic coordinate (base-pair position) of the CpG site on that chromosome.
`UCSC_CpG_Islands_Name`	Name/genomic range of the CpG island the site falls in, e.g. chr11:93861560-93862773.
`UCSC_RefGene_Group`	Position relative to the nearest gene, e.g. TSS200, TSS1500, 1stExon, 5'UTR, Body — useful for feature engineering, similar to how we used Gene region context in class.
`Relation_to_UCSC_CpG_Island`	Position relative to the CpG island itself: Island, N_Shore, S_Shore, N_Shelf, S_Shelf, or open sea (blank).
`Regulatory_Feature_Group`	Regulatory annotation if the site overlaps a known regulatory element, e.g. Promoter_Associated (often blank — treat missing as "none").
`Forward_Sequence`	~120-letter genomic sequence surrounding the CpG site, with the exact dinucleotide marked as [CG]. This is your short, motif-scale context — like the Sequence column in class, just centered on a real biological landmark.
`seq`	A longer raw DNA sequence (hundreds to thousands of letters) surrounding the site — your raw material for k-mer, GC-content, and Conv1D/LSTM/GRU features, exactly as in Section 8 of Week4_InClass.ipynb.
`Beta — TARGET`	DNA methylation level at that CpG site, ranging from 0 (fully unmethylated) to 1 (fully methylated). Present in train.csv only — this is what you are predicting for test.csv.

2. Your Task 🎯

Build a complete, end-to-end supervised learning pipeline — the same structure we practiced in class — on the DNA methylation dataset.

Suggested Problem Framing

(you may pick this OR propose your own, clearly justified)

Primary target: bin Beta into a 3-class Methylation_Level — Low (Beta < 0.3), Medium (0.3 – 0.7), High (Beta > 0.7) — and predict this class, mirroring exactly the Class_Label / Disease_Risk classification setup from Week4_InClass.ipynb. This keeps every technique from class directly reusable: k-mer features, Decision Tree / Random Forest / XGBoost, and CNN / LSTM / GRU on the raw sequence.

Alternative Target Ideas

(for groups who want a harder challenge)

- Binary classification: predict is_highly_methylated (Beta ≥ 0.7) vs. not, and discuss the class imbalance you encounter.
- Regression (bonus-level difficulty): predict the raw Beta value directly and evaluate with RMSE / R^2 instead of classification metrics.

Required Steps

(mirrors our in-class notebook)

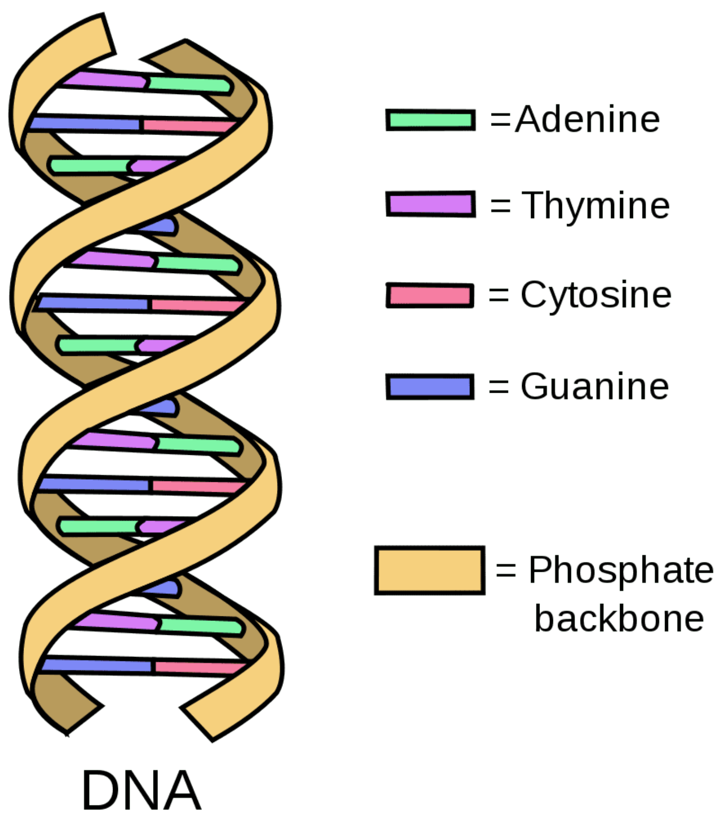
1. EDA & cleaning — check missing values (Regulatory_Feature_Group and CpG-island fields have many blanks), plot the distribution of Beta, and check whether chromosomes or gene regions are balanced.
2. Feature engineering — compute GC_Content, sequence length, a k-mer frequency vector, Purine/Pyrimidine ratio, and GC_Skew from seq / Forward_Sequence, exactly as in Section 4 of the in-class notebook.
3. Encoding — correctly choose One-Hot vs. Label encoding for UCSC_RefGene_Group, Relation_to_UCSC_CpG_Island, and Regulatory_Feature_Group, and justify your choice.
4. Stratified train/test split on your Methylation_Level classes (or a sensible split strategy if you chose regression).
5. Train at least 5 models — a mix of classical ML (Decision Tree, Random Forest, XGBoost) AND at least 2 PyTorch deep learning models trained directly on the raw sequence (CNN, LSTM, and/or GRU), following Section 8's encoding and architecture recipes.
6. Evaluate — with Accuracy, Precision, Recall, F1 (macro), and a Confusion Matrix; always compare against a majority-class baseline, exactly as we did in Section 7 & 9.
7. Discuss — for your chosen framing, which mistake matters more — missing a highly-methylated site or wrongly flagging one — and why that matters biologically.
8. Error analysis — look at a handful of misclassified CpG sites and hypothesize why the model got them wrong (e.g. ambiguous gene region, borderline Beta near a bin edge).
9. Conclusion — which model won, and would you trust it for real methylation-level screening? Why or why not?

3. Bonus Points 🌟

- Use an ML model we did NOT cover in class (e.g. LightGBM, CatBoost, k-NN, Naive Bayes) — +bonus
- Build the CNN + BiLSTM hybrid suggested at the end of the in-class notebook (feed the CNN's convolution output into an LSTM before pooling) — +bonus

- Engineer at least one feature not covered in class, such as CpG density in a sliding window or distance to the nearest TSS — +bonus
- Attempt the regression framing (predict raw Beta) in addition to your main classification model, and compare which framing is more useful in practice — +bonus

> 🏆 **Special Recognition Award**
>
> The top 3 students in the class with the highest F1 Score on this week's task will each receive **+1% extra credit** added to this week's assignment score. Report your best F1 Score clearly in your notebook's conclusion so it can be verified.



Good luck, and have fun exploring the epigenome! 🧬🚀